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Health-oriented routes for active mobility

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ABSTRACT

Introduction: Active modes of transport can be defined as means of travel and transport at reduced speed, which take up little space on public roads and do not emit air pollutants or noise, being seen as an economic, social, and environmental asset, and a real alternative to motorized vehicles. Given its characteristics, the environmental benefits retrieved from active modes of transport should not serve only as a justification for its promotion but as a decisive and active factor in the scope of its actual use, which translates into the provision of pleasant and less polluted routes. Thus, the main objective of this paper is the development of a model for planning and generating urban routes for active modes that have a less negative influence on active users (pedestrians and cyclists) regarding their exposure to air and noise pollution.

Methods: Conventional route planners are mostly designed to find the shortest, fastest, or less expensive route. In this research, the model to assign the best route will be presented for pedestrians and cyclists from an environmental point of view, by defining the least polluted, less noisy, and most health-friendly path. Based on the production of noise and air pollution maps, the model proposes the contamination of the network routes, which are used to determine health-oriented routes. The model is carried out through the calculation of the noise and pollution exposure for the routes found, compared to the shortest route.

Conclusion: It was possible to conclude that less polluted routes may imply a small increase in the length of routes, but a significant gain in terms of reduction in exposure to PM10 and noise for active modes. Thus, the use of health-friendly routes can be seen as a matter of public health and a way for promoting more sustainable mobility.

1. Introduction

The decrease in the use of the individual private vehicle implies a reduction in the emission of pollutants, noise levels, road collisions, and other economic problems related mainly to the congestion of the main roads and intersections of urban areas (Buekers et al., 2015; Rabl and de Nazelle, 2012; Ross et al., 2011). In addition, a wide collection of benefits ranging from environmental aspects to improvements in the functioning of the transport system can be achieved if walking and cycling are promoted (FHWA, 2006; Maizlish et al., 2017). These benefits can be mainly translated into a reduction in the emission of air pollutants and congestion of urban road networks (Arsenio and Ribeiro, 2015; Dias and Ribeiro, 2020; Hackl et al., 2019; Lovelace et al., 2017; Sælensminde, 2004).

Thus, this research work aims at developing a methodology to plan routes for active modes that will consider the type of environmental conditions that pedestrians and cyclists are subject to during their journeys in an urban environment, more specifically noise and air pollution. Hence, the best route for pedestrians and cyclists is assigned by generating the less noisy path, and the path

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